

New Wheels Project

Introduction to SQL

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
SELECT COUNT(DISTINCT o.customer_id) FROM order_t o;
```

```
SELECT c.state, COUNT(DISTINCT o.customer_id) AS num_customers FROM order_t o
JOIN customer_t c ON o.customer_id = c.customer_id GROUP BY c.state ORDER BY
num_customers DESC
```

Output:

Test Cases	Run SQL
Result: Passed	
Query 1	
Query:	
SELECT COUNT(DISTINCT o.customer_id) FROM order_t o	
Output:	
Showing 1 rows	
	COUNT(DISTINCT o.c...
	994

Test Cases	Run SQL
Result: Passed	
Query 1	
Query:	
SELECT c.state, COUNT(DISTINCT o.customer_id) AS num_customers FROM order_t o JOIN customer_t c ON o.customer_id = c.customer_id GROUP BY c.state ORDER BY num_customers DESC	
Output:	
Showing first 10 rows out of 49 rows	
state	num_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35

Observations and Insights:

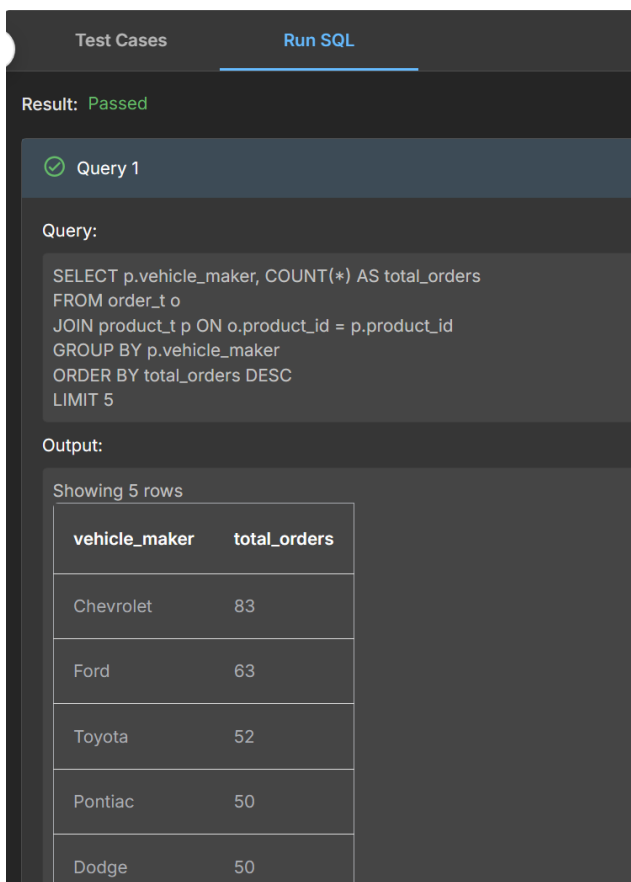
There are 994 in total of customers who have placed orders. Most of them are from Texas and California with 97 customers, followed by Florida with 86, New York 69 and District of Columbia with 35 customers.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker, COUNT(*) AS total_orders
FROM order_t o
JOIN product_t p ON o.product_id = p.product_id
GROUP BY p.vehicle_maker
ORDER BY total_orders DESC
LIMIT 5;
```

Output:



The screenshot shows a SQL query execution interface. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result is 'Passed'. A green checkmark icon is next to 'Query 1'. The query text is displayed in a dark box. Below the query, the output is shown as a table with 5 rows. The table has two columns: 'vehicle_maker' and 'total_orders'.

vehicle_maker	total_orders
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

The top 5 vehicles makers are Chevrolet with 83 orders, Ford with 63, Toyota with 52, Pontiac with 50 and Dodge with 50 orders.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, total_orders
FROM (
    SELECT c.state,
           p.vehicle_maker,
           COUNT(*) AS total_orders,
           RANK() OVER (PARTITION BY c.state ORDER BY COUNT(*) DESC) AS rnk
    FROM order_t o
    JOIN customer_t c ON o.customer_id = c.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) ranked
WHERE rnk = 1
ORDER BY state;
```

Output:

Test Cases

Run SQL

Query 1

Query:

```
SELECT state, vehicle_maker, total_orders
FROM (
    SELECT c.state,
           p.vehicle_maker,
           COUNT(*) AS total_orders,
           RANK() OVER (PARTITION BY c.state ORDER BY COUNT(*) DESC) AS rnk
    FROM order_t o
    JOIN customer_t c ON o.customer_id = c.customer_id
    JOIN product_t p ON o.product_id = p.product_id
    GROUP BY c.state, p.vehicle_maker
) ranked
WHERE rnk = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	total_orders
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3

Observations and Insights:

This query shows which vehicle maker is the most preferred in each state by giving how many orders are. For example, for Alabama it's Dodge with 5 total orders.

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
SELECT
    AVG(
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad'      THEN 2
            WHEN 'Okay'    THEN 3
            WHEN 'Good'     THEN 4
            WHEN 'Very Good' THEN 5
        END
    ) AS overall_avg_rating
FROM order_t;
```

```
SELECT
    quarter_number,
    AVG(
        CASE customer_feedback
            WHEN 'Very Bad' THEN 1
            WHEN 'Bad'      THEN 2
            WHEN 'Okay'    THEN 3
            WHEN 'Good'     THEN 4
            WHEN 'Very Good' THEN 5
        END
    ) AS avg_rating
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```



Output:

Output:

Showing 1 rows

overall_avg_rating
3.135

Output:

Showing 4 rows

quarter_number	avg_rating
1	3.554838709677419
2	3.354961832061069
3	2.9563318777292578
4	2.3969849246231156

Observations and Insights:

The overall average rating is 3.135.

For the first quarter the average is around 3.55, for the second quarter it is 3.35 on average, for the third quarter it is 2.96 on average, and for the fourth quarter it is 2.40 on average rating.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    customer_feedback,
    COUNT(*) * 100.0 / (SELECT COUNT(*) FROM order_t) AS percentage
FROM order_t
GROUP BY customer_feedback
ORDER BY percentage DESC;
```

```
SELECT
    quarter_number,
    COUNT(CASE WHEN customer_feedback IN ('Bad', 'Very Bad') THEN 1 END) * 100.0
    / COUNT(*) AS dissatisfaction_rate
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT
  customer_feedback,
  COUNT(*) * 100.0 / (SELECT COUNT(*) FROM order_t) AS percentage
FROM order_t
GROUP BY customer_feedback
ORDER BY percentage DESC
```

Output:

Showing 5 rows

customer_feedback	percentage
Very Good	22.6
Good	21.5
Okay	20.2
Bad	18.2
Very Bad	17.5

Test Cases

Run SQL

Result: Passed

✓ Query 1

Query:

```
SELECT
  quarter_number,
  COUNT(CASE WHEN customer_feedback IN ('Bad', 'Very Bad') THEN 1 END) * 100.0
  / COUNT(*) AS dissatisfaction_rate
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	dissatisfaction_rate
1	22.258064516129032
2	29.00763358778626
3	40.61135371179039
4	59.79899497487437

Observations and Insights:

Overall we see that the percentage of customer feedback is evenly distributed. There's not a category of customer feedback that has a significant difference with the other categories: very good is with 22.6%, good with 21.5, okay with 20.2, bad with 18.2 and very bad with 17.5.

We can see that the dissatisfaction rate created to run the query increases each quarter by ending with a percentage of around 60% dissatisfied by the fourth quarter.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Test Cases

Run SQL

Result: Passed

✓ Query 1

Query:

```
SELECT
    quarter_number,
    COUNT(*) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

We can see that over each quarter the number of total orders start to decrease. It starts with 310 in quarter 1, going with 262 in quarter 2, 229 orders in quarter 3 and in quarter 4 with 199 orders.

Question 7: Calculate the net revenue generated by the company.
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
SELECT
    SUM(vehicle_price * quantity * (1 - discount)) AS total_net_revenue
FROM order_t;
```

```
SELECT
    quarter_number,
    ROUND(SUM(vehicle_price * quantity * (1 - (discount / 100.0))), 2) AS
net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Test Cases

Run SQL

Result: Passed

✓ Query 1

Query:

SELECT
SUM(vehicle_price * quantity * (1 - discount)) AS total_net_revenue
FROM order_t

Output:

Showing 1 rows

total_net_revenue
48610993.78130001

Test Cases

Run SQL

Result: Passed

✓ Query 1

Query:

```
SELECT
  quarter_number,
  ROUND(SUM(vehicle_price * quantity * (1 - (discount / 100.0))), 2) AS net_revenue
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue
1	39421580.16
2	32715830.34
3	29229896.19
4	23346779.63

Observations and Insights:

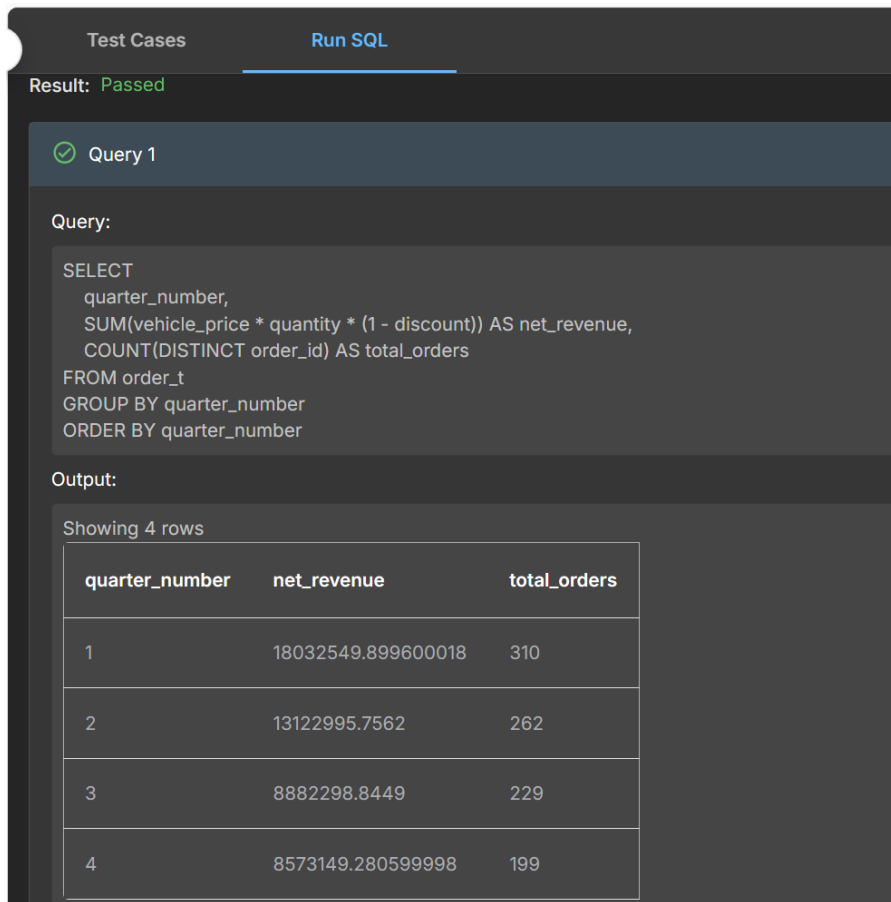
Net revenue declined consistently across the year, with quarter-over-quarter decreases of 17.03% in Q2, 10.67% in Q3, and 20.15% in Q4, indicating a downward sales trend.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT
    quarter_number,
    SUM(vehicle_price * quantity * (1 - discount)) AS net_revenue,
    COUNT(DISTINCT order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL execution environment with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Passed'. A green checkmark icon is next to 'Query 1'. The query text is displayed in a code editor, and the output is shown as a table with 4 rows.

Result: Passed

Query 1

Query:

```
SELECT
    quarter_number,
    SUM(vehicle_price * quantity * (1 - discount)) AS net_revenue,
    COUNT(DISTINCT order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	total_orders
1	18032549.899600018	310
2	13122995.7562	262
3	8882298.8449	229
4	8573149.280599998	199

Observations and Insights:

Both net revenue and orders decrease over each quarter. The largest revenue drop occurs between Q2 and Q3, while orders steadily decline each quarter, suggesting both sales volume and revenue per order are weakening over time.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount / 100), 4) AS avg_discount_percent
FROM order_t o
JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_percent DESC;
```

Note: The 'discount' field in the 'order_t' table is stored as a percentage, i.e. 0.6 represents a discount of 0.6%, not 60%.

Output:

Result: Passed

✓ Query 1

Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount / 100), 4) AS avg_discount_percent
FROM order_t o
JOIN customer_t c
    ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY avg_discount_percent DESC
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount_percent
laser	0.0064
mastercard	0.0063
visa-electron	0.0062
maestro	0.0062

Observations and Insights:

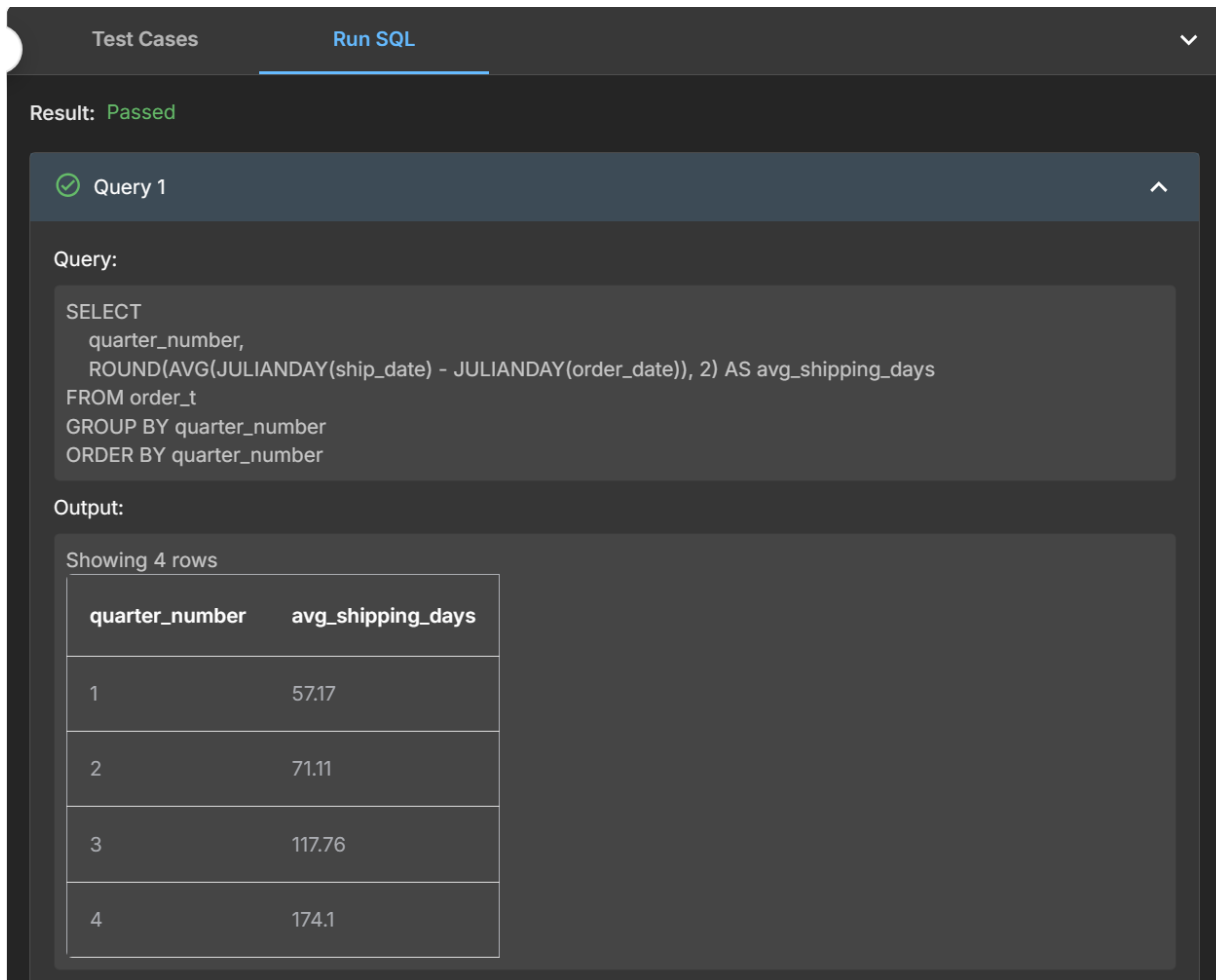
The average discount is consistent across all card types around 0.61% – 0.64%. Customers using Laser cards received the highest average discount (~0.64%). Customers using Switch, JCB, and Diners Club US/CA received the lowest average discounts (~0.61%). The differences are minor (just 0.03% of difference), suggesting discounts aren't strongly tied to card type.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS
avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL execution environment with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Result: Passed'. A section titled 'Query 1' contains the SQL query. Below the query, the output is displayed as a table with 4 rows.

Result: Passed

Query 1

Query:

```
SELECT
    quarter_number,
    ROUND(AVG(JULIANDAY(ship_date) - JULIANDAY(order_date)), 2) AS avg_shipping_days
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_shipping_days
1	57.17
2	71.11
3	117.76
4	174.1

Observations:

This shows that the average shipping days increased over each quarter. It took around 2 months in the first quarter, increasing to 6 months in the fourth quarter.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
48610993.78	1000	994	3.14
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
8573149.28	199	174.1	20.1

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions.

Business Recommendations

- Reduce Shipping Times:**
 The average shipping duration has increased from around 2 months in Q1 to around 6 months in Q4. This is a major driver of customer dissatisfaction. Improving logistics and partnering with more reliable shippers can lower this average shipping duration and increase customer satisfaction.
- Improve Customer Experience & Feedback:**
 Dissatisfaction grew steadily across quarters, ending with almost of 60% of customers reporting *Bad* or *Very Bad* feedback. Initiatives that improve customer experience such as having responsive customer support and proactive issue resolution can improve this.
- Boost Sales Performance:**
 Net revenue and total orders are both declining each quarter (–17.03% in Q2, –10.67% in Q3, –20.15% in Q4). Focused marketing campaigns, loyalty programs, and competitive pricing strategies are needed to retain and attract customers.

Other recommendations:

- Leverage popular vehicle makers such as Toyota and Ford while promoting these brands is a way to increase sales.
- Regional strategy: focus on campaigns where there's a larger customer base such as New York, Texas, California
- Maintain fair discounts.